

How Stale Can an Optical Digital Twin Be?

A Measured Freshness Budget for QoT Decisions

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Abstract: A GNPY-class digital twin makes format decisions between telemetry syncs while the network fills. We measure the failure curve: nine additions stay under a 5% risk target after a sparse sync; mature bands tolerate nearly unlimited fill.

1 Introduction

Digital twins of the optical physical layer are moving from research to practice [2]. A twin holds a QoT model of the deployed network, and operators use it to answer provisioning questions: which format will this lightpath sustain, how much margin does it need. The twin’s answer is only as good as its view of the network, and the view goes stale. Between telemetry syncs, channels are added, nonlinear interference grows, and the GSNR the twin believes is no longer the GSNR the receiver sees. The question every deployment must answer is how often the twin needs to sync. Today that answer is set by engineering judgment. This paper gives it a measured number.

We treat each twin decision as a certificate. The twin clears a format for a lightpath, and the certificate fails when the deployed channel cannot support that format. We call this failure a false-clear, and we measure its rate on the GN-model physical layer of GNPY [1] over the CORONET-CONUS reference network, with fill statistics drawn from a public demand matrix. Three results follow. The freshness budget is a property of the band’s maturity, not a constant: a twin synced on a sparse provisioning band shows no false-clears through four added channels, stays within a 5% risk target through nine, and fails from the tenth add onward, while a twin synced on a band of twenty or more channels tolerates tens of adds, and at forty, the full C-band, without a single false-clear. Which services are exposed when the twin is stale is set by spectral position and reach, so a per-lightpath view recovers real capacity over a uniform margin. And when the twin’s QoT model is a learned surrogate, training it on average error leaves it blind exactly at the FEC threshold where its decisions are made.

2 Method

Substrate. Lightpaths are transparent shortest-path routes of the CORONET-CONUS reference topology; per-channel GSNR comes from the GN model in GNPY [1]. Table 1 lists the parameters. The twin’s view is the band as last synced. The staleness variable N is the number of channels present in

Table 1: Substrate parameters

Topology	CORONET-CONUS (75 nodes; routes ≤ 2500 km, 2–31 spans)
Fibre	SSMF, 0.2 dB/km, $D = 17$ ps/nm/km, 80 km spans
Amplifiers	EDFA, NF 6 dB
Launch power	+2 dBm per channel, flat (not re-optimised per fill)
Grid	32 GBd on 50 GHz, C-band of 76 channels
QoT engine	GNPY GN model (analytic NLI)
Deployed fill	sampled from SNDlib janos-us-ca per-link distribution
Estimator noise	0.3 dB monitoring; 0.5 dB RMSE in §3.2

the C-band that co-propagates with the service along its own path (its interference neighbourhood), and ΔN counts the channels added to that band since the sync. N runs from a lightly filled provisioning band of 6 to a full grid of 76. The filling protocol is fixed and disclosed: the band fills as a contiguous comb of 32 GBd channels on the 50 GHz grid, from the low-frequency edge of the C-band (191.35 THz) upward, so an add always lands at the top of the comb. Each sweep point evaluates the certificate over 40 spectral positions per fill level, across the route pool’s span-count classes weighted by route multiplicity; the sweep advances one channel at a time through the crossing region and repeats with the twin synced at $N_0 \in \{6, 20, 40, 60\}$ channels. A certificate selects the highest of five declared formats whose required GSNR the twin’s estimate clears: QPSK, 8QAM, 16QAM, 32QAM and 64QAM at 6.5, 9.0, 12.5, 16.0 and 19.0 dB respectively (32 GBd, roughly 20%-overhead soft-decision FEC; declared values, no probabilistic shaping). The capacity result of §3.2 uses a rate-flexible transceiver instead: spectral efficiency $\min(6, \log_2(1 + \text{SNR}/\Gamma))$ per polarisation with a $\Gamma = 1$ dB gap for FEC and implementation.

Certificate, consumer, witness. A certificate selects the highest format whose required GSNR the twin’s estimate clears. The consumer is that decision. The witness is the true deployed GSNR, in practice the receiver’s pre-FEC BER. For the surrogate study the features are the synced GSNR plus spectral position, reach and fill, the label carries an irreducible term for unmeasured network state, and two gradient-boosted objectives compete: average-error (squared loss) and decision-

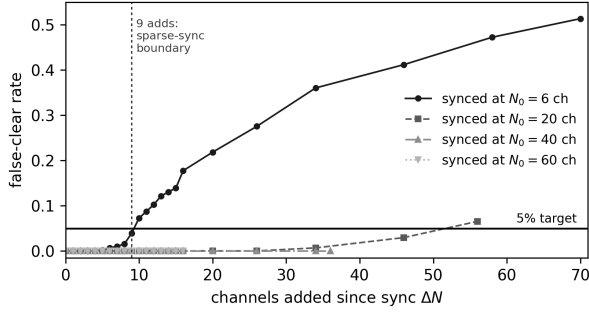


Figure 1: The freshness curve (GNPy over CORONET-CONUS, single-add sweep, deterministic). False-clear against channels added since sync, for four sync baselines N_0 . Sparse-synced certificates fail from the tenth add; mature-band syncs tolerate tens of adds ($N_0 = 20$) or the full band ($N_0 \geq 40$) without a false-clear. Marker shape distinguishes baselines.

aware (a conservative quantile loss). Bars and manipulation checks are pre-registered. The sealed record is content-hashed and graded on seeds disjoint from those used during design [7].

3 Results

3.1 The freshness budget

The risk of a stale twin is driven by the fill added since its last sync, and even more by how full the band already was at the sync. Advancing the deployed band one channel at a time while the twin’s view stays at the sync point gives the failure curve directly (Fig. 1), and we report the curve rather than a fit. Synced at the sparse provisioning fill of 6 channels, the certificate produced no false-clears through the first four adds in this evaluation, stays within a 5% risk target through nine adds, exceeds it from the tenth (7.3%), and saturates near 0.5 as the band fills. (An earlier draft quoted an interpolated 8.1-add crossing from a coarser grid and, before that, a linear law; the single-add sweep supersedes both.) The budget is strongly baseline-dependent, in the reassuring direction. Re-synced at 20 channels, the same certificate shows no false-clears through 26 further adds and crosses 5% only around fifty; synced at 40 or 60 channels it produces none at all, out to the full band. The mechanism is the shape of nonlinear interference growth: the first neighbours cost the most GSNR, so a twin synced on a sparse band certifies against a nearly best-case view and its decisions sit close to format thresholds, while a twin synced on a mature band has already absorbed most of the interference it will ever see. The operational rule follows, and it is not a calendar. The dangerous window is the young band: after provisioning at sparse fill, re-sync or re-certify within about nine adds. Once the band carries twenty or more channels, ordinary fill no longer breaks the certificate, and frequent re-syncs buy nothing.

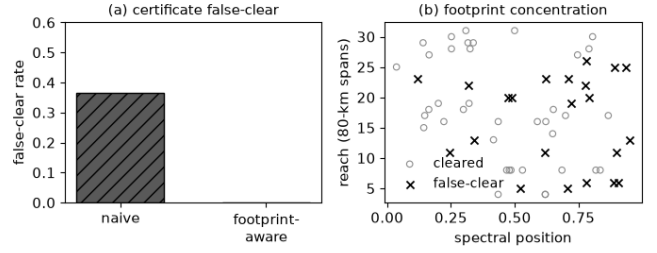


Figure 2: Where a stale twin fails (GNPy over CORONET-CONUS). Per-service false-clear over spectral position and reach concentrates on threshold-adjacent services. A per-lightpath view removes it. Marker shape distinguishes classes.

3.2 Who is exposed when the twin is stale

Staleness does not spread its damage evenly. A lightpath’s GSNR drops as neighbours arrive, and whether it crosses a format threshold depends on where it sits: false-clears concentrate on services whose spectral position and reach leave their GSNR near a boundary, and are absent elsewhere (Fig. 2). So a single blanket margin cannot protect the fleet efficiently. We price this against today’s practice of one uniform system margin, with fill statistics grounded on public demand data. Routing the 1482 demands of the SNDlib janos-us-ca reference matrix [6] (39 nodes, 61 undirected links) gives a per-link channel-fill distribution with median 10 and 90th percentile 35 of 76; each CORONET lightpath’s deployed fill is then sampled independently from that marginal distribution. The demand data grounds the fill statistics, not a routed traffic snapshot: correlations between links along a route are not carried over. What matters for the comparison is the heavy tail, which is exactly what a marginal preserves: a uniform margin sized for the few near-full links over-provisions the lightly filled majority. The two policies are specified fully. Each chooses the smallest margin on a 0.25 dB grid that holds the fleet false-clear at or below 0.01 under the same 0.5 dB Gaussian estimator error. The uniform policy provisions every lightpath at its provisioning-time GSNR minus one blanket margin, and the fleet constraint forces 2.75 dB. The footprint-aware policy evaluates each lightpath’s GSNR at its own deployed fill, interpolated between the provisioning and full-band GN anchors, and then needs only 1.25 dB. At that matched safety the aware policy recovers about 8% of spectral efficiency, near 0.33 b/sym or 21 Gb/s per channel, on a rate-flexible transceiver model. The gain is capacity the uniform margin strands on links that never fill.

3.3 The learned surrogate fails at the threshold

Many twins replace the physics model with a learned surrogate for speed. How that surrogate is trained matters more than how accurate it is on average. Uncertainty-aware and quantile QoT models are established [3]; what has not been measured is the price of ignoring them at the decision boundary. Under irreducible prediction uncertainty the average-error surrogate wins the metric it was trained on and loses the one the twin ex-

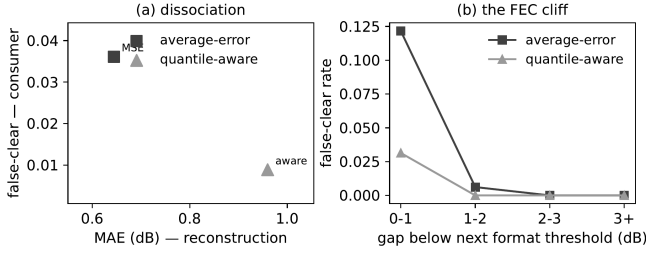


Figure 3: Two surrogate objectives, two metrics (GNPy, gradient-boosted regressors). (a) The average-error objective has lower MAE and higher false-clear than the quantile objective. (b) Its false-clears concentrate where the true GSNR sits within a decibel below the next format threshold; services already clearing the top format cannot false-clear and are excluded. Marker shape distinguishes objectives.

ists for. Its MAE is 0.65 dB against 0.95 dB for the quantile objective, yet its false-clear rate is about 0.041 against 0.008, five times higher, at comparable delivered spectral efficiency near 3.9 bits/symbol (Fig. 3a). The failures sit where the true GSNR lies within about a decibel below the *next* format’s threshold (Fig. 3b): there a small upward prediction error clears one format too high, and nowhere else can it. That is the cliff, and a symmetric error metric is blind there. A uniform backoff is the fair baseline and we report it without leakage: a backoff calibrated on a designated validation half of the sealed per-prediction records (0.90 dB, then frozen) matches the quantile model on the held-out half, false-clear 0.006 against 0.007 at equal delivered bits. So the dissociation does not require quantile training to fix; it requires *some* decision-aware calibration. The quantile objective (here $\tau = 0.15$) reaches that operating point from its training loss alone, with no separate calibration set, and the machinery already exists [3].

4 Related work

Optical digital twins are an active area, from GNPy-based twins fed by network telemetry [2] to calibration against field data; that work builds and tunes the twin, and does not quantify how stale it can afford to be. Margin erosion under channel add is known qualitatively, and impairment-aware provisioning accounts for it at setup time [5]; our contribution is the measured staleness law and the budget it implies for the twin that persists after setup. Uncertainty-aware ML-QoT, including quantile models, is established [3, 4]; we credit it as the correction for the surrogate result and add the measured cliff dissociation that motivates it. We do not offer a better twin, estimator or allocator. We offer the freshness budget, the exposure map, and the price of the wrong training objective.

5 Discussion

A digital twin is a certificate factory, and certificates age. The freshness measurement turns twin synchronization from a judgment call into a provisioned quantity, and the quantity

is a property of band maturity: re-sync a sparse-synced band within about nine adds at a 5% risk target, spend no sync effort on mature bands, protect the threshold-adjacent services first, and calibrate surrogates on the decision rather than the average. The evidence stands on the GN model with public demand data, and its limits are worth stating. Launch power is flat at +2 dBm and not re-optimised as the band fills, ROADM filtering penalties are not modelled, and the witness is the GN-model GSNR rather than a receiver’s pre-FEC BER. These effects do not all push the budget the same way: re-optimising launch power as the band fills would offset part of the drop the twin misses, while filtering penalties would add to it. The budget should be re-measured, not extrapolated, and a coherent-receiver witness on a testbed twin is the natural next step. In companion graded experiments, two margin policies with equal fleet budgets received opposite verdicts from two service classes whose failures ride different axes of the network state (cells XPROTO-QOT-FLIP and XPROTO-CSI-FLIP in the sealed repository [7]); a full treatment of that inversion is beyond this paper’s scope and is reported separately. The same consumer-relative failure appears in radio link adaptation and in database replication [7]. The results here stand on the optical QoT certificate alone.

References

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